Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/06/2020
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Individual Differences in the Neuronal and Behavioral Response to Exposure to HZE Particles		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
Start Date:	08/01/2016	**End Date:**	07/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 5/4/2020)		
Key Personnel Changes/Previous PI:	May 2020 report: CoInvestigator (CoI) Shibu Poulouse, Ph.D., is no longer CoI on the project.		
COI Name (Institution):	Shukitt-Hale, Barbara Ph.D. (US Department of Agriculture/Agricultural Research Service, Beltsville, MD)		
Grant/Contract No.:	NNX16AE06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Future missions in space (such as a mission to Mars or the establishment of a moon base) will involve long-term travel beyond the magnetic field of the Earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles, and particles of high energy and charge (HZE particles). Previous research has shown that exposure to single, above threshold doses of HZE particles and protons can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements or may produce degenerative changes in neuronal and behavioral functioning after the conclusion of the mission. The effects of exposure to multiple, subthreshold doses of the same or different particles or protons have not been extensively studied. The objectives of the proposed research program are: (1) to evaluate the effects of repeated below threshold/above threshold exposures to the same and different HZE particles and protons on cognitive and neuronal deficits and how they relate to the characteristics of the specific particle; (2) to determine the relationship between changes in performance and HZE particle- and proton-induced changes in oxidative stress, inflammation, signaling molecules, and autophagy in key brain regions; and (3) to determine how individual characteristics, including age and sex, may influence the responsiveness of the individual to HZE particle irradiation. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and protons and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Our data indicate that there is a significant risk that exposure to low, NASA-relevant doses of HZE particles produces alterations in central nervous system (CNS) function and in the behaviors that are dependent upon the integrity of specific CNS systems. However, current research suggests several caveats that might modulate the effectiveness of such exposures in disrupting neuronal function and cognitive performance. The experiments presented in this report were designed to evaluate several factors which may affect the degree to which exposure to space radiation might affect the cognitive ability of astronauts on exploratory class missions. The first series of experiments utilized exposure to 4He and 56Fe particles in male and female rats. The second series of experiments utilized exposure to 4He and 1H particles because these particles will constitute the major source of radiation within the spacecraft during exploratory class missions. The first series of experiments examined the effects of repeated exposure to a below threshold dose of 4He particles on a subsequent exposure to either a below or above threshold dose 56Fe particles separated by 72 hr on cognitive performance in male and female rats. The experimental design attempts to mimic the most likely pattern of radiation exposure to which astronauts will be exposed on exploratory class missions involving repeated exposures to below threshold doses. In this context, threshold is defined as the lowest dose of a specific particle which will produce a deficit in cognitive performance. The results indicated that the effects of combined exposures varied as a function of both the specific task (operant responding, novel object or novel spatial performance) and the sex of the subject. A second series of experiments evaluated the effects of exposure to 4He followed by exposure to 1H or exposure to 1H followed by exposure to 4He particles separated by 2 hrs. Because the majority of the dose experienced by astronauts within the spacecraft be provided by these particles, the order of irradiation was alternated; some rats received 4He particles 2 hr prior to exposure to 1H and some received1H prior to 4He particles. The effects of repeated exposures on cognitive performance were task dependent and also dependent upon the order of presentation. The third series of experiments rats was designed to determine whether the disruption of neuronal function and cognitive performance following exposure to 56Fe particles resulted from direct effects of the particles on neurons (“hits”) or whether the neuronal and cognitive effects were due to indirect effects of oxidative stress and neuroinflammation, which can be produced by the effects of HZE particles throughout the entire body. If oxidative stress and neuroinflammation are the critical mediators there should be no differences in the behavioral or neuronal endpoints as a function of the location of the exposure (head-only or body-only). If direct effects on neurons mediate the differences in the effects of higher or lower linear energy transfer (LET) particles, we should see threshold differences as a function of particle LET with head-only, but not with body-only exposures. The data indicates that irradiation-induced changes in neuronal function and the development of cognitive deficits are independent of location: head-only, body-only, or whole-body exposures produce similar changes in neuronal function and cognitive deficits.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Abstracts for Journals and Proceedings	Rabin BM, Rottman V, Jiang V, Hull M, Ingram I, Shukitt-Hale B. "Dietary modulation of the effects of exposure to HZE particles on cognitive performance." Presented at the 22020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Abstracts for Journals and Proceedings	Wyrobek AJ, Britten R, Rabin B, Bowles D, Rosi S, Witkowska HE, Fiehn O, Settles M, Froenicke L, Albou LP, Mungall C, Peterson LE, Straume T. "CNS and peripheral molecular predictors of individual risks for behavioral deficits and persistent neuropathies after exposure to space radiation." Presented at the 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Articles in Peer-reviewed Journals	Rabin BM, Miller MG, Larsen A, Spadafora C, Zolnerowich NN, Dell'Acqua LA, Shukitt-Hale B. "Effects of exposure to 12C and 4He particles on cognitive performance of intact and ovariectomized female rats." Life Sciences in Space Research. 2019 Aug;22:47-54. https://doi.org/10.1016/j.lssr.2019.07.005 ; PubMed PMID: 31421848 , Aug-2019
Articles in Peer-reviewed Journals	Davis CM, Rabin BM. "Space radiation-induced cognitive deficits following head-only, whole body, or body-only exposures." THREE. 2019 Sep 11. Review. https://three.jsc.nasa.gov/articles/Cog_Defects_Davis-Rabin.pdf , Sep-2019